

Technical Revision – Playwright with Python

Day 11–12: Locators & Interactions

Fixtures

- **page** → Represents a single tab.
- **context** → A browser context (isolated session with cookies, storage, etc.).
- **browser** → The browser instance (Chromium, Firefox, WebKit).

Locators

Locators in Playwright are strict, resilient **element selectors**.

Role-based (recommended, accessible, stable):

```
page.get_by_role("button", name="Login")
page.get_by_role("textbox", name="Username")
```

User-facing:

```
page.get_by_text("Submit")
page.get_by_label("Email")
page.get_by_placeholder("Search")
```

CSS/XPath:

```
page.locator("css=.btn-primary")
page.locator("xpath=//button[@id='submit']")
```

Interactions

Click:

```
page.get_by_role("button", name="Login").click()
```

Fill Input:

```
page.get_by_label("Username").fill("tomsmith")
```

Check/Uncheck:

```
page.locator("#agree").check()
```

```
page.locator("#agree").unchecked()
```

Dropdown:

```
page.locator("select#country").select_option("US")
```

Hover:

```
page.locator(".menu").hover()
```

Drag & Drop:

```
page.locator("#source").drag_to(page.locator("#target"))
```

Keyboard:

```
page.get_by_label("Search").press("Enter")
```

Day 13: Assertions & Waits

Web-First Assertions (expect)

Playwright's assertions are auto-waiting, retrying until the condition passes or timeout.

```
from playwright.sync_api import expect
```

Page Assertions

```
expect(page).to_have_title("Dashboard")  
expect(page).to_have_url("https://example.com/home")
```

Locator Assertions

```
locator = page.get_by_text("Welcome")
```

```
expect(locator).to_be_visible()  
expect(locator).to_be_enabled()  
expect(locator).to_be_checked()  
  
expect(locator).to_have_text("Welcome, Tom!")  
expect(locator).to_have_value("admin")  
expect(locator).to_have_attribute("src", "logo.png")
```

Count Assertions

```
items = page.locator(".list-item")  
expect(items).to_have_count(5)
```

Key Differences vs Traditional Selenium

- Playwright locators auto-wait for elements (no need for explicit waits).
- Assertions are web-first (retry until timeout).
- Rich set of user-facing locators (labels, roles, placeholders).
- Can combine UI + API testing in the same framework.